RATE ANALYSIS (2082/083)

Description of works:	Crusher Run Macadam Base and Sub Base,Providing and laying Crusher Run Macadam on a prepared surface, spreading and mixing , watering and compacting to form a layer of sub-base/Base course as per Drawing and Technical Specifications. (53mm maximum size)	Unit : 360 m3
Spec. cl. No: 1204		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
12.7A	Skilled	md	3.00	1215.00	3645.00	63mm to 45 mm	cu.m.	157.460	1990.00	313345.40	Grader	hr	6.00	3957.50	23745.00
	Unskilled	md	14.00	855.00	11970.00	22.5mm to 5.6mm	cu.m.	151.060	1990.00	300609.40	Vibaratory Roller	hr	6.00	2626.11	15756.66
						Below 5.6 mm	cu.m.	166.68	2044.00	340693.92					
						Cost of Water	ltr	36000	0.19	6930.00					
				Sub total of A =	15615.00				Sub total of B =	961578.72				Sub total of C =	39501.66
				Sub total of A +B + C =	1016695.38				Contractor's overhead expenses 15% =		152504.31			Unit Rate =	1169199.69
					Note :-For 53 mm maximum Size									Rate Per M3=	3247.78

Description of works:	Crusher Run Macadam Base and Sub Base,Providing and laying Crusher Run Macadam on a prepared surface, spreading and mixing , watering and compacting to form a layer of sub-base/Base course as per Drawing and Technical Specifications. (45mm maximum size)	Unit : 360 m3
Spec. cl. No: 1204		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
12.7A	Skilled	md	3.00	1215.00	3645.00	45 mm to 22.5mm	cu.m.	24.120	1990.00	47998.80	Grader	hr	6.00	3957.50	23745.00
	Unskilled	md	14.00	855.00	11970.00	22.4mm to 5.6mm	cu.m.	237.600	1990.00	472824.00	Vibaratory Roller	hr	6.00	2626.11	15756.66
						Below 5.6 mm	cu.m.	213.48	2044.00	436353.12					
						Cost of Water	ltr	36000	0.19	6930.00					
				Sub total of A =	15615.00				Sub total of B =	964105.92				Sub total of C =	39501.66
				Sub total of A +B + C =	1019222.58				Contractor's overhead expenses 15% =		152883.39			Unit Rate =	1172105.97
					Note :-For 45 mm maximum Size									Rate Per M3=	3255.85

Description of works:	Providing and applying prime coat with Hot Bitumen (including cutter) on prepared surface ofgranular base including cleaning of road surface and spraying by mechanical means as per Technical Specification .Prime Coat, with MC 30 / 70	Unit : 5000 lit.
Spec. cl. No: 1302		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.1 A	Skilled	md	3.00	1215.00	3645.00	Bitumen 60% MC30/MC70	kg	3150.00	110.00	346500.00	Mechanical broom	hr	8.00	2217.58	17740.64
	Unskilled	md	50.00	855.00	42750.00	cutback (kerosene) 40% MC30/MC70	kg	2100.00	130.97	275044.24	Air compresser	hr	8.00	1727.47	13819.76
											Bitumen Distributor	hr	6.00	4541.16	27246.96
											Boiler	hr	8.00	849.70	6797.60
											Generator	hr	8.00	1173.44	9387.52
				Sub total of A =	46395.00				Sub total of B =	621544.24				Sub total of C =	74992.48
				Sub total of A +B + C =	742931.72				Contractor's overhead expenses 15% =		111439.76			Unit Rate =	854371.48
														Rate Per Liter=	170.87

RATE ANALYSIS (2082/083)

Description of works:	Providing and applying prime coat with Hot Bitumen (including cutter) on prepared surface of granular base including cleaning of road surface and spraying by manual means as per Technical Specification .Prime Coat, with MC 30 / 70 (Manual)	Unit : 1000 lit.
Spec. cl. No: 1302		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.1 B	Skilled	md	3.00	1215.00	3645.00	Bitumen MC30/MC70	kg	660.00	110.00	72600.00	Tractor	hr	6.00	1283.38	7700.28
	Unskilled	md	100.00	855.00	85500.00	cutback (kerosene) 40% MC30/MC70	kg	440.00	130.97	57628.31	Bitumen Distributor	hr	6.00	4541.16	27246.96
											Boiler	hr	8.00	849.70	6797.60
											Generator	hr	8.00	1173.44	9387.52
				Sub total of A =	89145.00				Sub total of B =	130228.31				Sub total of C =	51132.36
				Sub total of A +B + C =	270505.67				Contractor's overhead expenses 15% =		40575.85			Unit Rate =	311081.52
														Rate Per Liter=	311.08

Description of works:	Providing and applying prime coat with Bitumen emulsion on prepared surface of granular base including cleaning of road surface and spraying primer at specified rate using mechanical means as per Technical Specification. Prime Coat, with Emulsion	Unit : 5000 lit.
Spec. cl. No: 1302		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.1 C	Skilled	md	3.00	1215.00	3645.00	Bitumen Emulsion	kg	5250.00	110.00	577500.00	Mechanical broom	hr	8.00	2217.58	17740.64
	Unskilled	md	40.00	855.00	34200.00	Water	lit	10000.00	0.19	1925.00	Air compresser	hr	8.00	1727.47	13819.76
											Emulsion Distributor	hr	6.00	4541.16	27246.96
				Sub total of A =	37845.00				Sub total of B =	579425.00				Sub total of C =	58807.36
				Sub total of A +B + C =	676077.36				Contractor's overhead expenses 15% =		101411.60			Unit Rate =	777488.96
														Rate Per Liter=	155.50

Description of works:	Providing and applying primer coat with Bitumen emulsion on prepared surface of granular base including cleaning of road surface and spraying primer at specified rate by manual means as per Technical Specification.Prime Coat, with Emulsion	Unit : 1000 lit.
Spec. cl. No: 1302		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.1 D	Skilled	md	3.00	1215.00	3645.00	Bitumen Emulsion	kg	1100.00	110.00	121000.00	Tractor-Trolley	hr	6.00	1316.38	7898.28
	Unskilled	md	80.00	855.00	68400.00	Water	lit	10000.00	0.19	1925.00	Emulsion Distributor	hr	6.00	4541.16	27246.96
				Sub total of A =	72045.00				Sub total of B =	122925.00				Sub total of C =	35145.24
				Sub total of A +B + C =	230115.24				Contractor's overhead expenses 15% =		34517.29			Unit Rate =	264632.53
														Rate Per Liter=	264.63

RATE ANALYSIS (2082/083)

Description of works:	Providing and applying tack coat with hot Bitumen at specified rate on the prepared non-bituminous surfaces including cleaning as per Technical Speciation . Tack coat with Bitumen By Mechanical Means	Unit : 5000 lit.
Spec. cl. No: 1302		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.2 A	Skilled	md	3.00	1215.00	3645.00	Bitumen (Paving Grade)	kg	5250.00	110.00	577500.00	Boiler	hr	6.00	849.70	5098.20
	Unskilled	md	20.00	855.00	17100.00						Air compresser	hr	6.00	1727.47	10364.82
											Bitumen Distributor	hr	6.00	4541.16	27246.96
											Generator	hr	6.00	1173.44	7040.64
				Sub total of A =	20745.00				Sub total of B =	577500.00				Sub total of C =	49750.62
				Sub total of A +B + C =	647995.62				Contractor's overhead expenses 15% =		97199.34			Unit Rate =	745194.96
														Rate Per Liter=	149.04

Description of works:	Providing and applying tack coat with hot Bitumen at specified rate on the prepared non-bituminous surfaces including cleaning as per Technical Speciation .Tack coat with Bitumen By Manual Means	Unit : 1000 lit.
Spec. cl. No: 1302		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.2 B	Skilled	md	3.00	1215.00	3645.00	Bitumen (Paving Grade)	kg	1100.00	110.00	121000.00	Boiler	hr	8.00	849.70	6797.60
	Unskilled	md	40.00	855.00	34200.00						Tractor	hr	6.00	1283.38	7700.28
											Bitumen Distributor	hr	6.00	4541.16	27246.96
											Generator	hr	8.00	1173.44	9387.52
				Sub total of A =	37845.00				Sub total of B =	121000.00				Sub total of C =	51132.36
				Sub total of A +B + C =	209977.36				Contractor's overhead expenses 15% =		31496.60			Unit Rate =	241473.96
														Rate Per Liter=	241.47

Description of works:	Providing and laying surface dressing as wearing course in single coat using gravel on a recently applied layer of bituminous binder on prepared surface as per Drawing andTechnical Specifications. Surface Dressing by mechanical means 19 mm Nominal Size Chipping	Unit : 6000 Sqm
Spec. cl. No: 1303		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.7 A ii	Skilled	md	3.00	1215.00	3645.00	Crushed Stone chipping	cum	102.0000	1506.00	153612.00	Roller (Pneumatic)	hr	12.00	3618.67	43424.04
	Unskilled	md	12.00	855.00	10260.00						Chip Spreader	hr	6.00	5040.23	30241.38
				Sub total of A =	13905.00				Sub total of B =	153612.00				Sub total of C =	73665.42
				Sub total of A +B + C =	241182.42				Contractor's overhead expenses 15% =		36177.36			Unit Rate =	277359.78
														Rate Per Sqm=	46.23

Description of works:	Providing and laying surface dressing as wearing course in single coat using gravel on a recently applied layer of bituminous binder on prepared surface as per Drawing andTechnical Specifications. Surface Dressing by mechanical means 13 mm Nominal Size Chipping	Unit : 7500 Sqm
Spec. cl. No: 1303		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.7 A ii	Skilled	md	3.00	1215.00	3645.00	Crushed Stone chipping	cum	87.0000	1506.00	131022.00	Roller (Pneumatic)	hr	12.00	3618.67	43424.04
	Unskilled	md	12.00	855.00	10260.00				130.97		Chip Spreader	hr	6.00	5040.23	30241.38
				Sub total of A =	13905.00				Sub total of B =	131022.00				Sub total of C =	73665.42
				Sub total of A +B + C =	218592.42				Contractor's overhead expenses 15% =		32788.86			Unit Rate =	251381.28
														Rate Per Sqm=	33.52

RATE ANALYSIS (2082/083)

Description of works:	Providing and laying surface dressing as wearing course in single coat using gravel on a recently applied layer of bituminous binder on prepared surface as per Drawing andTechnical Specifications. Surface Dressing by mechanical means 10 mm Nominal Size Chipping	Unit : 9000 Sqm
Spec. cl. No: 1303		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.7 A iii	Skilled	md	3.00	1215.00	3645.00	Crushed Stone chipping	cum	80.30	1614.00	129604.20	Roller (Pneumatic)	hr	12.00	3618.67	43424.04
	Unskilled	md	12.00	855.00	10260.00						Chip Spreader	hr	6.00	5040.23	30241.38
				Sub total of A =	13905.00				Sub total of B =	129604.20				Sub total of C =	73665.42
				Sub total of A +B + C =	217174.62				Contractor's overhead expenses 15% =		32576.19			Unit Rate =	249750.81
														Rate Per Sqm=	27.75

Description of works:	Providing and Laying open graded premix carpet of 20 mm thickness composed of 13.2 mm to 5.6 mm aggregates as wearing coarse on a previously prepared base as per drawing and technical specification. Mechanical Method using hot mix plant	Unit : 4000 Sqm.
Spec. cl. No: 1311		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.9B	Skilled	md	5.00	1215.00	6075.00	Bitumen (Paving Grade)	kg	5840.00	110.00	642400.00	Hot Mix Plant	hr	6.00	5690.02	34140.12
	Unskilled	md	16.00	855.00	13680.00	Crushed Stone Chipping	Cum	108.00	1614.00	174312.00	Paver	hr	6.00	5275.05	31650.30
											Roller	hr	6.00	1875.20	11251.20
											Generator	hr	6.00	1173.44	7040.64
				Sub total of A =	19755.00				Sub total of B =	816712.00				Sub total of C =	84082.26
				Sub total of A +B + C =	920549.26				Contractor's overhead expenses 15% =		138082.39			Unit Rate =	1058631.65
														Rate Per Sqm	264.66

Description of works:	Providing and Laying open graded premix carpet of 20 mm thickness composed of 13.2 mm to 5.6 mm aggregates as wearing coarse on a previously prepared base as per drawing and technical specification. Manual Method	Unit :500 Sqm.
Spec. cl. No: 1311		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.9A	Skilled	md	8.00	1215.00	9720.00	Bitumen (Paving Grade)	kg	730.00	110.00	80300.00	Mixer	hr	4.00	826.26	3305.04
	Unskilled	md	21.00	855.00	17955.00	Crushed Stone Chipping	Cum	13.50	1614.00	21789.00	Boiler	hr	4.00	849.70	3398.80
											Roller	hr	2.00	1875.20	3750.40
				Sub total of A =	27675.00				Sub total of B =	102089.00				Sub total of C =	10454.24
				Sub total of A +B + C =	140218.24				Contractor's overhead expenses 15% =		21032.74			Unit Rate =	161250.98
														Rate Per Sqm	322.50

Description of works:	Providing and laying of 20 mm thickness composed of 11.2 mm to 0.09 mm (Type A) or 13.2 mm to 0.09 mm (Type B) aggregates using bitumen as wearing course on a previously prepared base , including mixing in a suitable plant as per drawing and technical specifications(Close graded)	Unit :10250 sqm
Spec. cl. No: 1310		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.1 (Type A)	Skilled	md	6.00	1215.00	7290.00	Bitumen (Paving Grade)	kg	22500.00	110.00	2475000.00	Batch mix HMP	hr	6.00	1999.99	11999.94
	Unskilled	md	16.00	855.00	13680.00	11.2-0.09 mm aggregates	Cum	276.75	1614.00	446674.50	Paver Finisher	hr	6.00	5275.05	31650.30
											Generator	hr	6.00	1173.44	7040.64
											Smooth Wheel Roller	hr	6.00	1875.20	11251.20
				Sub total of A =	20970.00				Sub total of B =	2921674.50				Sub total of C =	61942.08
				Sub total of A +B + C =	3004586.58				Contractor's overhead expenses 15% =		450687.99			Unit Rate =	3455274.57
														Rate Per sqm	337.10

RATE ANALYSIS (2082/083)

Description of works:	Providing and Laying Bituminous Concrete/Asphat Concrete using crushed aggreagate of specified grading, premix with bituminous binder and filler as per drawing and technical specification.	Unit :95.5 Cum
Spec. cl. No: 1309		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.6A	Skilled	md	5.00	1215.00	6075.00	Bitumen (Paving Grade)	kg	12940.00	110.00	1423400.00	Batch mix HMP	hr	6.00	1999.99	11999.94
	Unskilled	md	15.00	855.00	12825.00	20 -10 mm aggregates	Cum	49.48	1940.00	95991.20	Paver Finisher	hr	6.00	5275.05	31650.30
						10-5 mm aggregated	Cum	32.52	1940.00	63088.80	Generator	hr	6.00	1173.44	7040.64
						5 mm and below aggregates	Cum	56.55	1940.00	109707.00	Smooth Wheel Roller	hr	12.00	1875.20	22502.40
						Filler	ton	2.83	1344.64	3805.33	Pneumatic Roller	hr	6.00	3618.67	21712.02
				Sub total of A =	18900.00				Sub total of B =	1695992.33				Sub total of C =	94905.30
				Sub total of A +B + C =	1809797.63				Contractor's overhead expenses 15% =		271469.64			Unit Rate =	2081267.27
														Rate Per Cum	21793.37

Description of works:	Providing and laying seal coat sealing the voids in a bituminous surface as per drawing and technical specification	Unit : 7858 Sqm
Spec. cl. No: 1310		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.11	Skilled	md	1.00	1215.00	1215.00	Crushed Stone chipping	cum	47.16	1614.00	76116.24	Roller (Pneumatic)	hr	6.00	3618.67	21712.02
	Unskilled	md	6.00	855.00	5130.00	bitumih		5340.0000	110.00	587400.00	Paver Finisher	hr	6.00	5275.05	31650.30
											Generator	hr	3.00	1173.44	3520.32
											HMP	hr	3.00	1999.99	5999 97
				Sub total of A =	6345.00				Sub total of B =	663516.24				Sub total of C =	62882.61
				Sub total of A +B + C =	732743.85				Contractor's overhead expenses 15% =		109911.58			Unit Rate =	842655.43
														Rate Per Sqm=	107.24

Description of works:	Providing and Laying Slurry Seal Consisting of a mixture of fine aggreagate, portland cement filler, bituminous emulsion and water on road surface including cleanin of surface mixing of slurry seal in suitable mobile plant, laying and compacting to provide even riding surface as per drawing and technical specification. 3mm thickness	Unit : 20000 Sqm
Spec. cl. No: 1310		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
13.12	Skilled	md	1.00	1215.00	1215.00	Fine aggreegate	cum	74.80	2329.00	174209.20	Mechanical Broom	hr	6.00	2217.58	13305.48
	Unskilled	md	7.00	855.00	5985.00	Binder/Bitumin	Kg	17160.00	110.00	1887600.00	Aircompressor	hr	6.00	1727.47	10364.82
						Filler/Cement	ton	2.64	1344.64	3549.85	Mobile Sullry Seal Equipment	hr	6.00	2565.95	15395.70
						Cost of Water	Ltr	12000.00	0.19	2310.00					0.00
				Sub total of A =	7200.00				Sub total of B =	2067669.05				Sub total of C =	39066.00
				Sub total of A +B + C =	2113935.05				Contractor's overhead expenses 15% =		317090.26			Unit Rate =	2431025.31
														Rate Per Sqm=	121.55

Description of works:	Providing and laying of hand pack Stone soling and packing with smaller stone on prepared surface as per Drawing and Technical Specifications.	Unit : 1 m3
Spec. cl. No: 1006		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
C10(6.5)	Skilled	md	0.00	1215.00	0.00	Stone	m^3	1.20	2218.00	2661.60					
	Unskilled	md	1.50	855.00	1282.50										